

What can we do?

Option A

Build a new operational semantics

- Scheduling behaviours becomes more complex
- Capturing all orders is difficult
- We may be too permissive or restrictive again

Option B

Build an axiomatic concurrency model

- Take inspiration from weak memory models
- Designed to reason about optimisation
- Separate intentional and incidental causality



An Axiomatic Concurrency Model

```
when(x) {   
  x.balance += 10;  
  when(x) {  }  
}
```

Build around candidate executions

Executions are the events of the program and their order